

Lab7 RLC Resonant Circuit

Table No. _____ Name & Stu No. _____ Score _____ of 10

Apprartus:

- | | |
|---|-----------------------|
| 1. Power Amplifier | 5. Capacitors |
| 2. Breadboard and jumper wires | 6. Oscilloscope |
| 3. Resistors ($510\ \Omega \times 2$) | 7. Function Generator |
| 4. Inductor (100mH) | 8. Digital Multimeter |

Part One:

- Use the DMM to measure the values of the components in Tables and note down the values in Tables. Also, measure the resistance R_L of the inductors. Use the measured values in all your calculations. _____/5pt

Table 1: Measured Values

	Nominal Value	Measured Value
R1	$510\ \Omega$	
R2	$255\ \Omega$	
R_L of 100mH inductor		
C	$0.5\ \mu\text{F}$	
C	$0.25\ \mu\text{F}$	

- Construct the circuit shown in Figure 2 using the breadboard with jumper-wires, here C is $0.5\ \mu\text{F}$, L is 100mH and R is $510\ \Omega$.

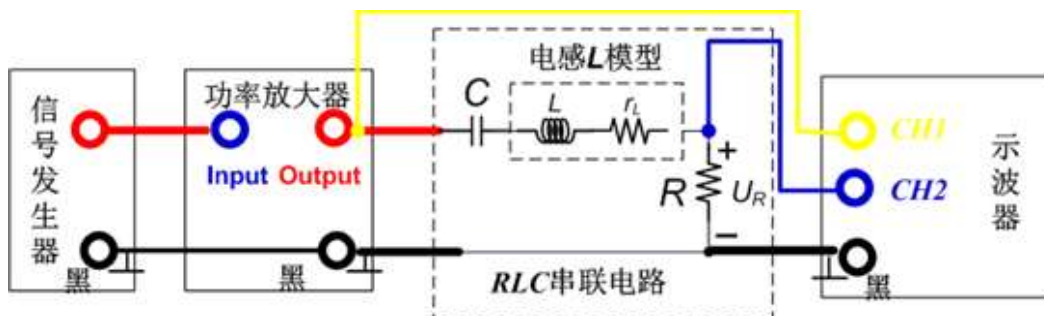


Fig.2 schematic diagram of RLC series circuit test

As shown in Figure 2, gain of the power amplifier is set to 1. Ch1 and CH2 of oscilloscope are used to observe the input voltage (that is, output voltage of power amplifier) and the voltage across the resistor U_R respectively. Make sure that the resistor is the **last component closest to the ground end of the circuit**.

- Use the Signal Generator to generate a Sine Wave with a 1.5V_{rms} amplitude. Set the frequency of the sin signal to be about the theoretic resonant frequency.
- Adjust the frequency of the signal generator slowly. When adjusting the frequency of the output signal of the signal generator, the output voltage of the power amplifier should be kept constant at 1.5V_{rms} . When the voltage at both ends of the resistor is the maximum

obtained from the oscilloscope, the frequency value at this time is the resonant frequency of the circuit. At this time, switch the YT mode of the oscilloscope to XY mode, and verify whether Lissajous curve is a straight line with the X axis (indicating TA to check signature)

- (3) Measure U_L (RMS Voltage) and U_C (RMS Voltage) under resonance state, and fill in the blank. **Note that the oscilloscopes only allow you to measure potential difference with respect to ground.** ___/6pt

C=0.5 μ F L=100 mH R=510 Ω	Measurement data				Calculation	
	f_0 (Hz)	U_R (V)	U_L (V)	U_C (V)	$I=U_R/R$ (mA)	$Q=U_C/U_i$

- (4) Measure the amplitude frequency characteristic curve of the RLC series circuit and complete the following table. Among them, f_1 and f_2 are the upper and lower frequencies of the passband FBW respectively. After f_0 and corresponding U_{R0} are measured, $U_{f1} = U_{f2} = 0.707 U_{R0}$ is calculated, and then f_1 and f_2 are measured from the values of U_{f1} and U_{f2} . ___/18.5pt

C=0.5 μ F L=100 mH R=510 Ω	Measurement data		Calculation	
	f (Hz)	U_R (V)	$I=U_R/R$ (mA)	I/I_0
	$f_1=$			0.707
	$f_0=$	$U_R=U_{R0}=$	$I=I_0=$	1
	$f_2=$			0.707

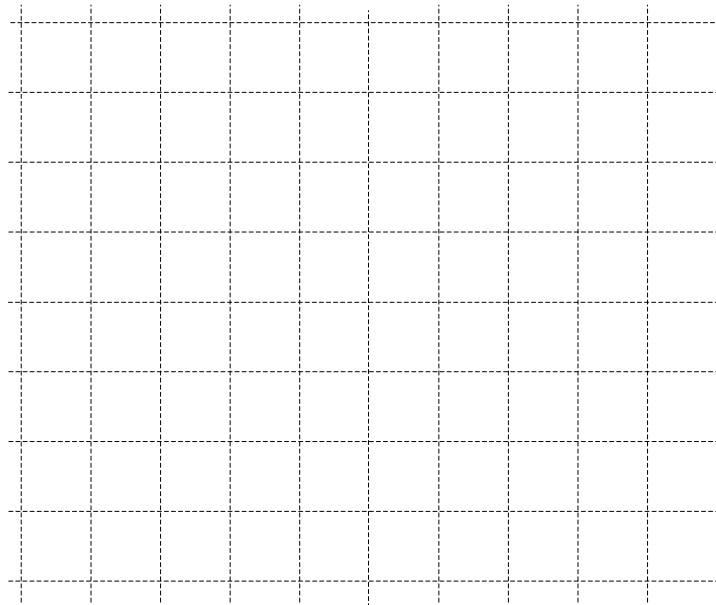
3. Change the resistor $R = 255\Omega$, Other parameters of the circuit remain unchanged. Repeat the above experimental steps, complete the following table. ___/24.5pt

C=0.5μF L=100mH R=255Ω	Measurement data				Calculation	
	<i>f</i> ₀ (Hz)	U _R (V)	U _L (V)	U _C (V)	I=U _R /R(mA)	Q=U _C /U _i
C=0.5μF L=100mH R=255Ω	Measurement data			Calculation		
	<i>f</i> (Hz)	U _R (V)		I=U _R /R (mA)		I/I ₀
	<i>f</i> ₁ =					0.707
	<i>f</i> ₀ =	U _R =U _{R0} =		I=I ₀ =		1
	<i>f</i> ₂ =					0.707

4. Change the capacitor $C=0.25\mu\text{F}$, Other parameters of the circuit remain unchanged, that is $L=100\text{mH}$, $R=255\Omega$. Repeat the above experimental steps. Under the resonant state, complete the following table. ___/11pt

C=0.25μF L=100mH R=255Ω	Measurement data				Calculation	
	f_0 (Hz)	U_R (V)	U_L (V)	U_C (V)	$I=U_R/R$ (mA)	$Q=U_C/U_i$
	Measurement data			Calculation		
	f (Hz)	U_R (V)		$I=U_R/（mA）$		I/I_0
	$f_1=$					0.707
	$f_0=$	$U_R=U_{R0}=$		$I=I_0=$		1
	$f_2=$					0.707

5. According to the experimental data, draw the frequency characteristic curve between I / I_0 and frequency f on the coordinate paper, and please use different colors to show the difference. Note: key information needs to be indicated. If the square is not enough, it can be expanded. ___/7pt



6. According to the three frequency characteristic curve in step5, calculate the Bandwidth (need to write the process). ___/6pt

C=0.5μF L=100mH R=510Ω	
C=0.5μF L=100 mH R=255Ω	
C=0.25μF L=100mH R=255Ω	

7. Calculate Q according to $Q = \frac{U_{C0}}{U_i}$ and $Q = \frac{\sqrt{L/C}}{R_{eq}}$ (show the calculation process) ___/9pt

Note: R_{eq} is the equivalent resistance seen from the input terminals. In this experimental circuit, except for the fixed resistance R , the inductor and capacitor all have a certain resistance value, so the total equivalent resistance is:

$$R_{eq} = \frac{U_i}{I_0}$$

C=0.5 μ F L=100mH R=510 Ω	
C=0.5 μ F L=100 mH R=255 Ω	
C=0.25 μ F L=100mH R=255 Ω	

8. Compare the calculated and the experimental resonant frequency, is the measurement larger or smaller than the calculated value? How much is the deviation? Analyze the possible reasons. ___/7pt

9. This experiment has been concerned with the series RLC circuit driven by a sinusoidal voltage source. Qualitatively discuss how the parameters affect the resonant frequency. ___/6pt